BAN 674 **Machine Learning** **Fall 2026**

Homework Assignment 1

The datasets are available as part of the book. Please register at https://www.dataminingbook.com/editions/python-2nd-edition-june-2025 to get access to the datasets.

1. Use ToyotaCorolla dataset. Prepare the dataset by doing the following:
    a. Summarize the dataset. (Do not just put the R/Python output. Make observations based on numbers)
    b. Normalize the variable kilometers
    c. Create dummies for the variable Fuel Type (see the script file in class for an example).
    d. Partition the data into three sets (see the script file in class for an example).

2. Market Street Wine case: Read the case and download the dataset and be ready to explore this dataset (just the training one) next week. Write a brief summary of the dataset to show me you have done this!

Please submit one pdf file and one R-script/Python file per team through Canvas. Your pdf file should contain the R/Python output and observations (if any). Please do not e-mail me your submission. E-mail submissions will not be accepted.